

Kyle Andrew Green

Senior Data Engineer

Naperville, IL 60565

Kylegreen.w11@gmail.com

(331) 333 1456

Senior Data Engineer with 12 years of experience designing and optimizing large-scale data platforms across cloud and on-prem environments. Expert in distributed data processing, real-time streaming, data modeling, orchestration and modern data warehousing. Strong background in Spark, Python, SQL, Airflow, Kafka, GCP, AWS and Azure. Experienced in retail, banking and healthcare domains, delivering solutions for personalization, recommendations, segmentation, promotions and transactional analytics. Passionate about understanding business use cases and building scalable data systems that drive measurable impact.

Skills

Languages

- Python, SQL, Scala, R, Shell Scripting

Cloud Platforms

- AWS (Redshift, Lambda, S3, Glue, EMR)
- Azure (Data Factory, Synapse, Azure ML)
- GCP (BigQuery, Dataflow, Pub/Sub)

Data Engineering

- ETL/ELT, Spark, Kafka, Hive, Sqoop, Flume, Apache Beam, Airflow

DevOps & Infra

- Kubernetes, Docker, Terraform, Jenkins, GitHub Actions

Monitoring & Logging

- CloudWatch, Prometheus, Grafana, ELK Stack, Datadog

Databases

- MongoDB, PostgreSQL, Oracle, Cassandra, SQL/NoSQL Systems

Machine Learning & AI

- MLOps, DVC, Model Deployment, Pandas, Scikit-Learn, PyTorch, TFX

Data Modeling

- Star Schema, Snowflake Schema, Dimensional Modeling, Data Vault

Data Warehousing

- Snowflake, BigQuery, Redshift, Synapse Analytics

Visualization

- Power BI, Tableau, Looker, Seaborn, Matplotlib

Testing

- Great Expectations, Unit Testing for Data Pipelines

Streaming

- Apache Flink, Kafka, Amazon Kinesis

Additional

- Data Governance, Git, Apache Superset, Web Scraping

Work Experience

Infinite Computer Solutions – New York, NY

Senior Data Engineer (Jun 2021 – Current)

- Engineered robust ETL/ELT pipelines using Apache Spark, Airflow and Python to process structured and semi-structured data from operational and third-party sources.
- Built real-time data ingestion pipelines with Apache Kafka, Kafka Connect and Flink to support low-latency business intelligence and predictive analytics.
- Integrated data from on-premise and cloud-native systems into AWS Redshift, BigQuery and Snowflake to enable unified analytics across environments.
- Designed and deployed infrastructure-as-code using Terraform to automate provisioning of Kubernetes clusters, data services and monitoring stacks.

- Implemented Airflow DAGs with built-in data quality validations using Great Expectations to ensure trust and integrity in mission-critical pipelines.
- Automated model deployment workflows using DVC, MLOps best practices and Azure ML for continuous training and inference in production.
- Developed dimensional data models using Star and Snowflake schema methodologies for optimized reporting and self-service analytics.
- Transformed raw and semi-structured data using PySpark and SQL into structured datasets ready for downstream consumption and ML feature engineering.
- Created streaming architectures using Amazon Kinesis and Apache Flink for fraud detection and near-real-time customer behavior tracking.
- Deployed containerized data services and ML applications using Docker and Kubernetes, ensuring reproducibility and scalability across dev and prod.
- Built centralized logging and alerting systems using ELK Stack, CloudWatch and Grafana to enhance pipeline observability and uptime.
- Conducted performance tuning on Redshift and BigQuery queries, reducing latency and optimizing costs for large-scale analytics workloads.
- Collaborated with cross-functional teams including product managers, data scientists and business analysts to align data solutions with strategic goals.
- Managed version control and CI/CD workflows using GitHub Actions and Jenkins for seamless integration and delivery of data applications.
- Designed and implemented data governance practices, including schema registry integration and access controls, to comply with regulatory requirements.
- Mentored junior engineers and conducted regular code reviews to uphold engineering best practices and foster team growth.
- Led migration efforts from legacy ETL systems to modern, cloud-native architectures using GCP Dataflow and Apache Beam.
- Delivered actionable dashboards and KPIs using Power BI, Looker and Tableau for executive and operational decision-making.
- Conducted R&D on scalable machine learning pipelines using TensorFlow Extended (TFX) and PyTorch for experimentation and deployment.
- Collaborated with data stewards and compliance teams to establish metadata management and lineage tracking for critical datasets.
- Played a key role in technical hiring, onboarding and establishing knowledge-sharing practices across the data engineering team.

Kroger - Cincinnati, OH

Data engineer (Sep 2016 - Mar 2021)

- Developed high-throughput real-time data pipelines using Apache Kafka, AWS Kinesis and Spark Streaming, capable of handling large-scale event ingestion for operational analytics.
- Built a data lake architecture on AWS (S3, Glue, Athena) to centralize and manage large-scale datasets, reducing data retrieval latency and improving accessibility.
- Engineered and maintained scalable Snowflake data warehouses, improving query efficiency through clustering, micro-partitioning and materialized views.
- Designed secure authentication and role-based authorization systems to manage sensitive data access across multiple data services.
- Automated CI/CD pipelines using GitHub Actions, enhancing deployment velocity and reducing manual errors in testing and production releases.
- Integrated AWS and Azure service operators for seamless deployment from GitHub to Kubernetes, streamlining continuous delivery processes.
- Constructed a distributed ETL framework with Apache Spark and Airflow, processing large daily batch workloads and improving overall job execution speed.
- Applied XSD and XSLT standards to transform and validate XML data, enhancing data integrity and compatibility in cross-platform environments.
- Built and exposed RESTful and GraphQL APIs to support integration with internal data services and analytics platforms.
- Improved pipeline reliability by implementing automated unit and integration testing, increasing test coverage and reducing failures.
- Performed performance testing using JMeter to identify bottlenecks in data services and implemented tuning strategies to reduce request latency.
- Customized internal data workflows and integrations using Microsoft Dynamics data APIs to support enterprise data operations.
- Performed component-level testing using Jest, Jasmine and Karma for internal data-driven UI tools supporting

analytics teams.

- Conducted performance testing and optimization for internal data applications to improve responsiveness and throughput.

Rivell - Sewell, NJ

Software developer (Feb 2013 - May 2016)

- Led modernization of deployment processes using Kubernetes for container orchestration and CI/CD pipelines through Jenkins and GitLab, improving delivery of backend data services.
- Built internal analytics tools using Python and Django, integrating secure real-time communication with backend data systems.
- Implemented Elasticsearch-based search solutions with data synchronization pipelines from PostgreSQL for high-performance geospatial and text search.
- Designed and developed RESTful APIs using Python (Django + DRF) and MongoDB to support scalable data-driven applications.
- Configured and optimized uWSGI servers to handle high-concurrency data workloads and backend service communication.
- Utilized Scrapy to extract, normalize and ingest external data sources into internal analytics pipelines.
- Standardized Python and JavaScript development practices using linters and code quality tools to improve maintainability.
- Implemented API-driven content delivery workflows integrated with internal CMS systems for data-rich applications.
- Collaborated in agile sprints to design, develop and unit test backend data modules and services.
- Demonstrated strong understanding of cloud-native design patterns, secure development and REST API best practices across backend systems.
- Facilitated transition of backend systems to modular, cloud-optimized architectures, improving deployment frequency and system resiliency.
- Provided technical mentorship on backend engineering and DevOps practices within cross-functional teams.
- Ensured secure and performant API interactions across distributed systems, applying security guidelines and OAuth2 standards.
- Maintained strong emphasis on code review, TDD and documentation to uphold engineering excellence and compliance.

Education

Bachelor's Degree: Computer Science

Lewis University | Romeoville, IL
August 2008 – May 2012